Chapter 1

Factoring Polynomials

We factor a polynomial into two or more polynomials of lower degree. For example, let's say that you wanted to factor $5x^3 - 45x$. You would note that you can factor out $5x$ from every term. Thus,

$$5x^3 - 45x = (5x)(x^2 - 9)$$

You might notice that the second factor looks like the difference of squares, so

$$5x^3 - 45x = (5x)(x + 3)(x - 3)$$

That is as far as we can factorize this polynomial.

Why do we care? The factors make it easy to find the roots of the polynomial. This polynomial evaluates to zero if and only if at least one of the factors is zero. Here, we see that

- The factor $(5x)$ is zero when x is zero.
- The factor $(x + 3)$ is zero when x is -3.
- The factor $(x - 3)$ is zero when x is 3.

So, looking at the factorization, you can see that $5x^3 - 45x$ is zero when x is 0, -3, or 3.

This is a graph of that polynomial with its roots circled:

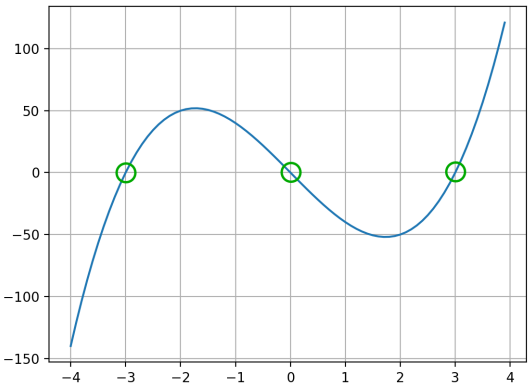


Figure 1.1: Factoring for roots makes it easier to find the roots rather than in expanded form.

1.1 How to factor polynomials

The first step when you are trying to factor a polynomial is to find the greatest common divisor for all the terms, then pull that out. In this case, the greatest common divisor will also be a monomial. Its degree is the least of the degrees of the terms, its coefficient will be the greatest common divisor of the coefficients of the terms.

For example, what can you pull out of this polynomial?

$$12x^{100} + 30x^{31} + 42x^{17}$$

The greatest common divisor of the coefficients (12, 30, and 42) is 6. The least of the degrees of terms (100, 31, and 17) is 17. So, you can pull out $6x^{17}$:

$$12x^{100} + 30x^{31} + 42x^{17} = (6x^{17})(2x^{83} + 5x^{14} + 7)$$

Exercise 1 Factoring out the GCD monomial

Factor out the greatest common divisor monomial from each polynomial.

Working Space

1. $18x^3 - 15x$
2. $4x^3 - 12x^2 + 16x$
3. $18x^{50} - 24x^{20} + 30x^8$

Answer on Page 11

So, now you have the product of a monomial and a polynomial. If you are lucky, the polynomial part looks familiar, like the difference of squares or a row from Pascal's triangle.

Often, you are trying to factor a quadratic like $x^2 + 5x + 6$ in a pair of binomials. In this case, the result would be $(x + 3)(x + 2)$. Let's check that:

$$(x + 3)(x + 2) = (x)(x) + (3)(x) + (2)(x) + (3)(2) = x^2 + 5x + 6$$

Notice that 3 and 2 multiply to 6 and add to 5. If you were trying to factor $x^2 + 5x + 6$, you would ask yourself, "What are two numbers that when multiplied equal 6 and when added equal 5?" And you might guess wrong a couple of times. For example, you might say to yourself, "Well, 6 times 1 is 6. Maybe those work. But 6 and 1 add 7. So those don't work."

Solving these sorts of problems are like solving a Sudoku puzzle. You try things and realize they are wrong, so you backtrack and try something else.

The numbers are sometimes negative. For example, $x^2+3x-10$ factors into $(x+5)(x-2)$.

Exercise 2 Factoring quadratics

Factor each quadratic into a pair of binomials.

Working Space

1. $x^2+14x+33$
2. $x^2-16x+60$
3. $x^2+3x-54$
4. $x^2-3x-10$

Answer on Page 11

Computing in Python

Python can check a factorization for you. Start by writing the original polynomial and its factored form as two Python functions:

```
def p(x):
    return 5*x**3 - 45*x

def f(x):
    return (5*x)*(x + 3)*(x - 3)
```

In Python, `**` means "raised to the power of," and every multiplication needs a `*`. You have to write `5*x`, since `5x` is an error.

First, confirm the roots we read off the factors:

```
for x in [0, -3, 3, 1]:
    print(x, p(x))
```

```
0 0
```

```
-3 0
3 0
1 -40
```

The values 0, -3, and 3 all give zero, so they are roots. The value 1 gives -40, so it is not a root.

Next, check that `p` and `f` are the same polynomial. If the factorization is correct, the two functions give the same output for every x:

```
for x in [0, 1, 2, 3]:
    print(x, p(x), f(x), p(x) == f(x))
```

```
0 0 0 True
1 -40 -40 True
2 -50 -50 True
3 0 0 True
```

Each line shows x, the value of the original, the value of the factored form, and whether they match. Every line says `True`.

How many values of x do you need to check? Here is a useful fact: if two polynomials of degree at most n give the same value at $n + 1$ different values of x, they are the same polynomial. Our polynomial has degree 3, so four matching values are enough.

Matching at the roots alone can hide a mistake. Suppose you forgot the 5 when you pulled out $5x$:

```
def g(x):
    return x*(x + 3)*(x - 3)

for x in [0, -3, 3, 1]:
    print(x, p(x), g(x), p(x) == g(x))
```

```
0 0 0 True
-3 0 0 True
3 0 0 True
1 -40 -8 False
```

The wrong factorization has the same roots, so it passes at 0, -3, and 3. The value $x = 1$ catches the mistake.

Exercise 3 Predict the output

Without running the code, predict what this prints. Which of the two values of x is a root?

```
1 def q(x):
2     return x**2 + 3*x - 4
3
4 print(q(2))
5 print(q(-4))
```

Working Space

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Exercise 4 Spot the sign error

A student claims that $x^2+3x-10=(x-5)(x+2)$ and runs this check:

```
def a(x):
    return x**2 + 3*x - 10

def b(x):
    return (x - 5)*(x + 2)

for x in [0, 1, 2]:
    print(x, a(x), b(x), a(x) ==
    ↪  b(x))
```

The output is:

```
0 -10 -10 True
1 -6 -12 False
2 0 -12 False
```

The first line says `True`. Why is the factorization still wrong? What is the correct factorization?

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Exercise 5 Spot the bug

A student wants to check that $x^2 + 5x + 6 = (x + 3)(x + 2)$, which we already know is correct, and writes:

```
1  def c(x):
2      return x^2 + 5*x + 6
3
4  def d(x):
5      return (x + 3)*(x + 2)
6
7  for x in [0, 1, 2]:
8      print(x, c(x), d(x), c(x) ==
       ↪  d(x))
```

The output is:

```
1  0 8 6 False
2  1 12 12 True
3  2 16 20 False
```

The factorization is correct, so what went wrong?

Working Space

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Exercise 6 Factor, check, and find the roots

Factor $2x^3 - 6x^2 - 8x$ completely. Write the original polynomial and your factored form as two Python functions, and check that they match at enough values of x. Then use your factors to list the roots, and confirm that each one gives zero.

Working Space

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This last part is optional. It uses SymPy, a Python library for algebra. SymPy is a separate install, so you may need to run `python3 -m pip install sympy` first.

```
from sympy import symbols, factor, expand

x = symbols('x')
print(factor(5*x**3 - 45*x))
print(expand(5*x*(x + 3)*(x - 3)))
```

```
5*x*(x - 3)*(x + 3)
5*x**3 - 45*x
```

The line `x = symbols('x')` makes `x` a *symbolic variable*: a letter that stands for any number, which is how x works in algebra. Earlier in this section, `x` always held one number at a time. Here it holds no number at all.

The function `factor` factors a polynomial, and `expand` multiplies the factors back out. SymPy lists $(x - 3)$ before $(x + 3)$; the order of the factors does not change the product.

SymPy also tells you when a polynomial cannot be factored using integers:

```
print(factor(x**2 + 1))
```

```
x**2 + 1
```

SymPy returns the polynomial unchanged, so x^2+1 has no factors with integer coefficients.

SymPy can factor every polynomial in this chapter in one line, which means it can also skip the thinking this chapter is teaching. Use it after you factor by hand, to check your answer.

If you get stuck on a problem, open `factor_hint.py`, type your polynomial on the line that starts with `polynomial =`, and run the file. It first shows a hint: the polynomial with its greatest common factor pulled out, found with SymPy's `factor_terms`. Try to finish the problem from there. When you press Enter, the file shows the full factorization.

Exercise 7 Predict what SymPy prints

Without running the code, predict the output.

Working Space

```
1 from sympy import symbols, factor
2
3 x = symbols('x')
4 print(factor(x**2 - 9))
```

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Exercise 8 Factor by hand, then check with SymPy

Factor $3x^2+12x-15$ completely by hand. Then use `expand` to multiply your factors back out, and use `factor` to compare your answer with SymPy's.

Working Space

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Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 2)

1. The GCD of 18 and 15 is 3, and the least degree is 1, so $18x^3 - 15x = (3x)(6x^2 - 5)$.
2. The GCD of 4, 12, and 16 is 4, and the least degree is 1, so $4x^3 - 12x^2 + 16x = (4x)(x^2 - 3x + 4)$.
3. The GCD of 18, 24, and 30 is 6, and the least degree is 8, so $18x^{50} - 24x^{20} + 30x^8 = (6x^8)(3x^{42} - 4x^{12} + 5)$.

Answer to Exercise 2 (on page 3)

1. 3 and 11 multiply to 33 and add to 14, so $x^2 + 14x + 33 = (x + 3)(x + 11)$.
2. -6 and -10 multiply to 60 and add to -16, so $x^2 - 16x + 60 = (x - 6)(x - 10)$.
3. 9 and -6 multiply to -54 and add to 3, so $x^2 + 3x - 54 = (x + 9)(x - 6)$.
4. -5 and 2 multiply to -10 and add to -3, so $x^2 - 3x - 10 = (x - 5)(x + 2)$.

Answer to Exercise 3 (on page 5)

```
6
0
```

At $x = 2$: $4 + 6 - 4 = 6$. At $x = -4$: $16 - 12 - 4 = 0$. So -4 is a root and 2 is not. This matches the factorization $x^2 + 3x - 4 = (x + 4)(x - 1)$.

Answer to Exercise 4 (on page 6)

At $x = 0$ both sides equal the constant term, -10, because -5 and 2 do multiply to -10.

A single match proves little here. The polynomial has degree 2, so all three values must match, and two of them fail. The numbers -5 and 2 add to -3 instead of 3, so the signs are swapped. The correct factorization is $(x+5)(x-2)$.

Answer to Exercise 5 (on page 7)

In Python, `^` does not mean "to the power of." It is a different operator, so `c` computes something other than x^2+5x+6. The fix is to write `x**2`. Notice that the bad version still matches at $x=1$ by chance, which is another reason to check several values.

Answer to Exercise 6 (on page 7)

Pull out $2x$ to get $(2x)(x^2-3x-4)$. The numbers -4 and 1 multiply to -4 and add to -3, so $2x^3-6x^2-8x=(2x)(x+1)(x-4)$.

```
def e(x):
    return 2*x**3 - 6*x**2 - 8*x

def h(x):
    return (2*x)*(x + 1)*(x - 4)

for x in [0, 1, 2, 3]:
    print(x, e(x), h(x), e(x) == h(x))

for r in [0, -1, 4]:
    print(r, e(r))
```

The polynomial has degree 3, so four values are enough, and all four lines say `True`. The roots are 0, -1, and 4, and each one prints 0.

Answer to Exercise 7 (on page 9)

```
(x - 3)*(x + 3)
```

The polynomial x^2-9 is a difference of squares, so it factors into $(x-3)(x+3)$.

Answer to Exercise 8 (on page 9)

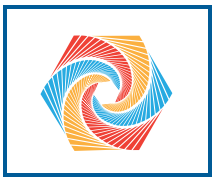
Pull out 3 to get $3(x^2 + 4x - 5)$. The numbers 5 and -1 multiply to -5 and add to 4, so $3x^2 + 12x - 15 = 3(x + 5)(x - 1)$.

```
from sympy import symbols, factor, expand

x = symbols('x')
print(expand(3*(x + 5)*(x - 1)))
print(factor(3*x**2 + 12*x - 15))
```

```
3*x**2 + 12*x - 15
3*(x - 1)*(x + 5)
```

Expanding gives back the original polynomial, and SymPy's factors match yours with the binomials in a different order.



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